

Nightingale

Technical Brief

STATUS	BASELINE	LAST UPDATED
Prototype - verified with limits	HEAD 51a1ce4 + F Final working tree	03 Sep 2026
SCOPE	Synthetic-data prototype for a shared longitudinal care record; not production medical software.	

1. Abstract

Nightingale is a shared longitudinal care record that turns fragmented dated notes into one verifiable patient story. Real-world Events remain the Timeline unit; parallel Artifacts preserve raw, AI-scribed, clinician, staff, and patient-facing representations without overwriting one another. Glance surfaces current priorities while every derived statement resolves to an exact source Span.

2. Design principles

- Timeline = what happened. Glance = what matters now. Patient View = what the patient needs to know or do.
- Server-side role, clinic, and patient scope is authoritative; the frontend is never the security boundary.
- AI assists language only. Provider output cannot create authority, bypass safety, or become its own source.

3. Proposed architecture

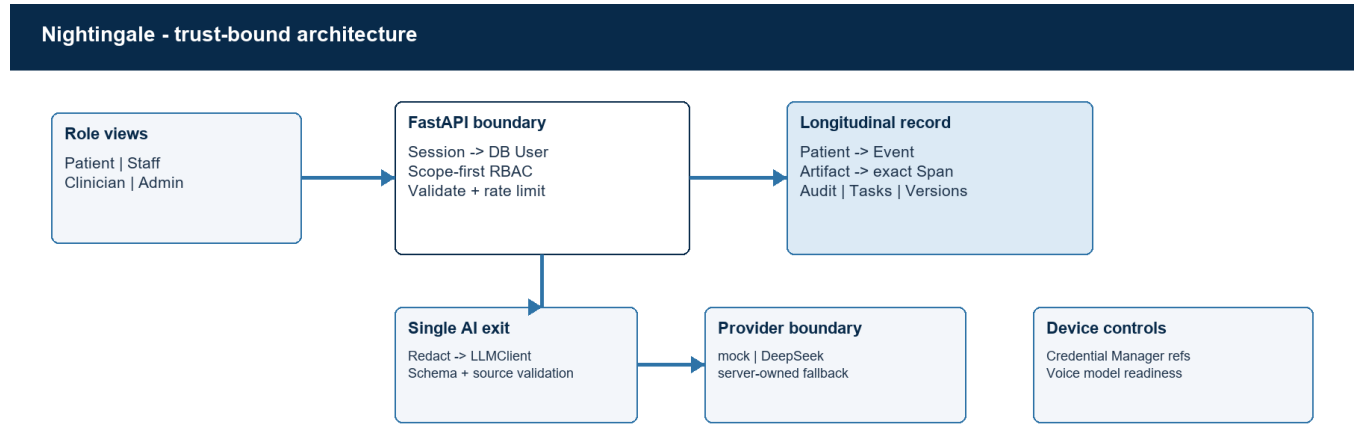


Figure 1. Trust boundaries, durable record, and the single AI egress path.

Authority boundary	The server owns RBAC, safety, state, Task authority, provenance, and limits. AI/patient-derived content never overwrites clinician or staff Artifacts.
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4. Data model and provenance

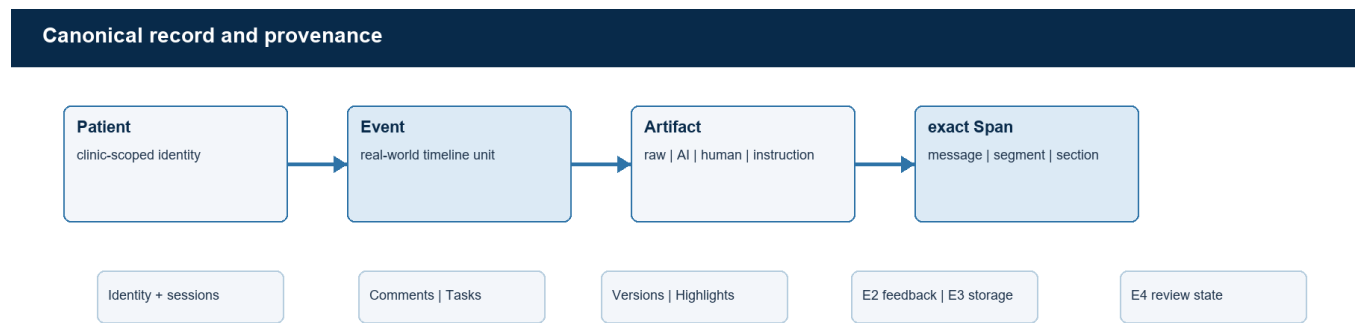


Figure 2. Candidate Brief Entry maps to Event + parallel Artifacts; exact Span is the trust anchor.

5. Role journeys

Role	Primary safe journey
Clinician	Read Glance in under 10 seconds -> inspect exact source -> review Event lifecycle and collaboration.
Staff	Add same-clinic Staff Notes, Nurse Consults, comments, and Tasks without assuming clinician authority.
Patient	Read instructions and Tasks -> complete a bounded raw-first Check-in -> return/correct/confirm safely.

6. RBAC and authority

Role	Can	Cannot
Patient	Own safe view, visible Tasks, own Check-in	Internal notes/comments; other patient; clinical authoring
Staff	Same-clinic staff notes, Nurse Consult, Tasks	Clinician authorship; other clinic
Clinician	Same-clinic review, notes, Copilot	Staff authorship; other clinic
Admin	Identity/session/invite and device settings	Clinical Note, Copilot, Task, or plan authoring

7. Patient Multi-turn Check-in

active	awaiting confirmation	submitted	safety escalated	abandoned
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Every patient message is committed first with a stable message_id. At most four clarification questions are allowed. Deterministic safety and medical-request rules run after raw persistence; confirmation creates the formal AI Summary, and every candidate quote must resolve to one patient message or be dropped.

8. Verification status

Capability	Status	Evidence boundary
E1 Role workspaces	IMPLEMENTED_AND_VERIFIED	Role-specific UI, RBAC, scope, and state isolation.
E2 + SL1-SL3 ranking	IMPLEMENTED_WITH_LIMITS	Formal Glance is deterministic base-only; frozen synthetic Shadow models and the observed-feedback bridge do not serve a model.
E3 Data-decay shadow	IMPLEMENTED_AND_VERIFIED	Protection-first shadow payload; not total DB savings.
E4 Voice + consult review	IMPLEMENTED_WITH_LIMITS	Default-off local lifecycle; Doctor confirmation requires three source-review attestations. Real ASR inputs and multilingual clinical validity remain NOT_RUN.
Patient Multi-turn Check-in	IMPLEMENTED_AND_VERIFIED	Raw-first, bounded safety, concurrency, RBAC, exact provenance.
E5 Submission package	REPOSITORY PACKAGE COMPLETE	Repository artifacts exist; external video playback, link access and email delivery remain owner-controlled evidence.

9. Measured evidence

Signal	Observed	Boundary
Backend	710 passed / 2 skipped	712 collected; skips require ignored local ASR model/audio inputs. F Final 9/9 passed.
Frontend	Build passed / 112 modules	TypeScript no-emit and Vite production build passed.
Focused regression	75 safety + 62 learning	Provenance/RBAC/publication/concurrency and SL1/SL2/SL3 Shadow isolation passed.
Glance	Layer B P95 5.115 ms	100 local samples after 10 warm-ups; target <= 300 ms.

10. Provider, ASR, learning, and security boundary

- Check-in mock and deterministic fallback journeys are verified; DeepSeek live turn succeeded, but the strict final Summary failed validation and fell back.
- D3 live Provider layer is NOT_RUN by design; D4 frozen Copilot evaluation passed with provider=mock.
- E4 is default-off local faster-whisper with no diarization or invented confidence. Doctor Consult confirmation now requires three source-review attestations. A synthetic Malay-English-Hokkien statement passed exact transport and provenance tests; translation, physical microphone, clinical multilingual understanding, and noisy accuracy are NOT_RUN.
- Formal Glance remains deterministic base-only. SL1 records all candidate decisions, SL2 uses frozen synthetic role models in Shadow, and SL3 has no eligible real-feedback model. Names, IC/ID patterns, and phones are redacted before LLM egress; SQLCipher, TLS, metadata-only audit, and Credential Manager references protect the local prototype.

11. Trade-offs

Decision	Why
Deterministic ranking before LLM ranking	Transparent, fast, reproducible, and auditable under a 72-hour prototype constraint.
Shadow learning behind deterministic controls	Complete candidate snapshots, fixed bands, protection, provenance, caps, freeze/replay and rollback permit evaluation without automatic unsafe demotion or a serving claim.